Assay for Transposase-Accessible Chromatin with Sequencing:

Materials and Equipment:

- 20 Days old tomato, or older.
- Sterile 100mm x 20 mm Petri plates (CELL STAR)
- Razor Blades
- Buffers Listed at the end

Method: (Important Note: All pipette tips used for nuclei should be cut off before pipetting).

1. Add 1ml of ATAC lysis Buffer in a petri-dish then collect fresh tomato leaves tissue around 0.5 grams and place them above the buffer as shown in Figure 1.

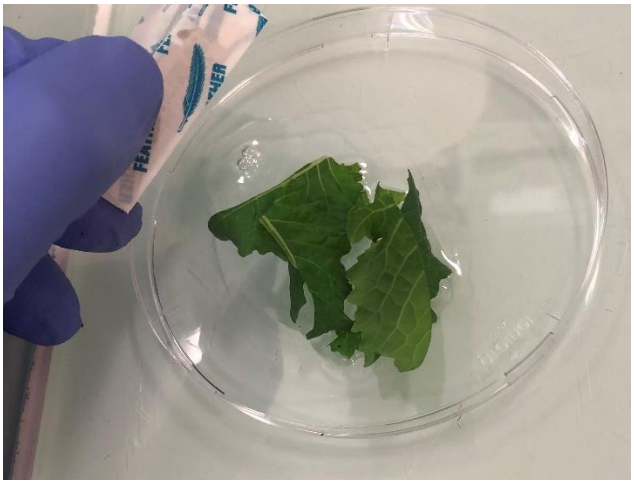


Figure 1. Placement of tomato leaves on the ATAC lysis buffer

2. Finely chop tomato leaves with razor blade as shown in Figure 2.

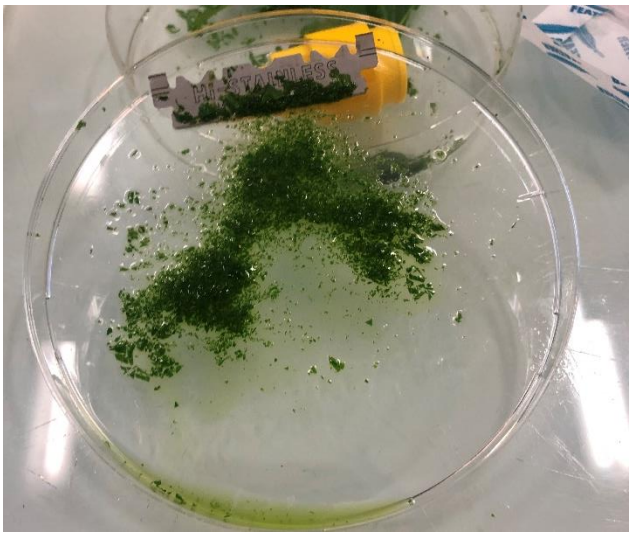


Figure 2. Finely chopped tomato leaves in ATAC lysis buffer.

3. Add 500 ul of ATAC lysis buffer and cut the tip of the 1ml pipette tip and wash the dish well as shown in Figure 3.

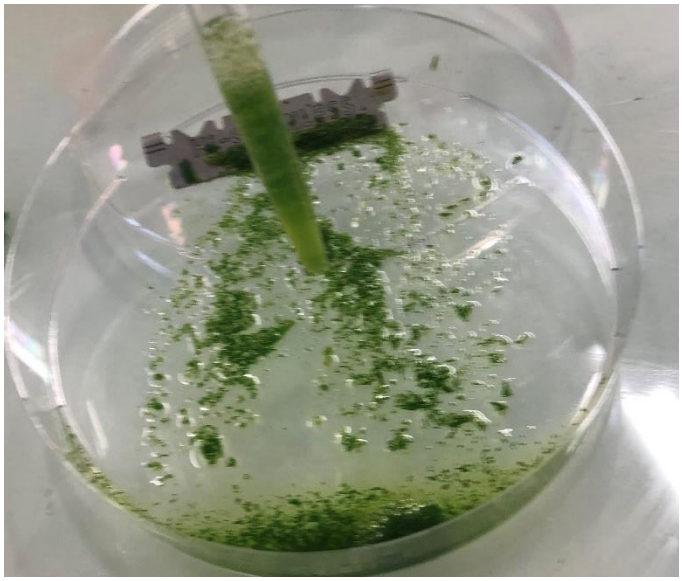


Figure 3. 1ml pipette tip cut and washing the petri-dish well

4. Transfer sample into 50 um filter in a 1.5ml tube with the cut tip pipette as shown in Figure 4.



Figure 4. 50 um filter placed on 1.5ml tube.

5. Pellet nuclei at 600g for 5 min at 4°C.
6. Discard supernatant and resuspend in 1ml ATAC wash buffer.
7. Pellet nuclei at 600g for 5 min at 4°C.
8. Discard supernatant and resuspend in 1ml ATAC wash buffer.
9. Pellet nuclei at 600g for 5 min at 4°C. (if the pellet still looks green, proceed for one more round of washing before going to step 10)
10. Resuspend pellet in 1ml ATAC wash buffer, and count nuclei with flow cytometer as shown an example in figure 5.

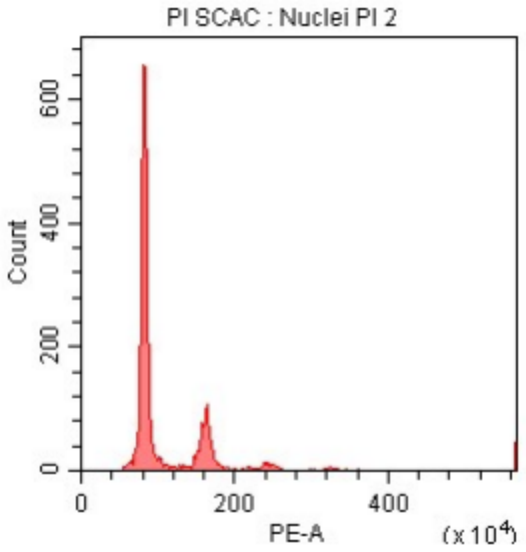
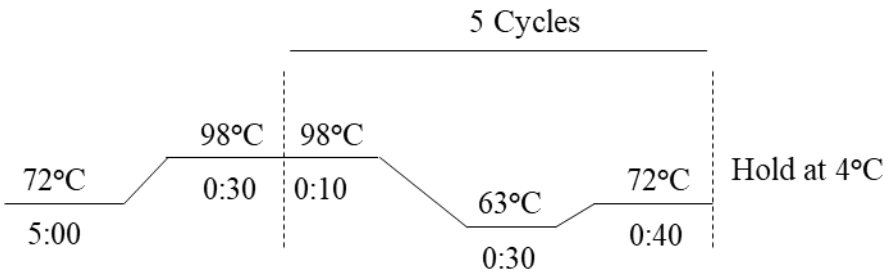
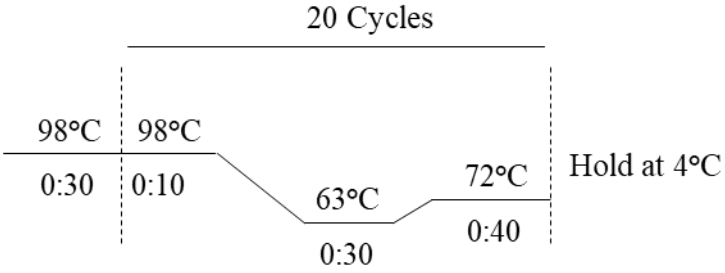


Figure 5. Counts of different tomato nuclei in flow cytometer using propidium Iodide stain.

11. Transfer 50,000 nuclei into a new 1.5 ml tube, (cryopreserve the rest by adding them to DMSO solution (Final Concentration 10%) and place them in cryopreservation container at -80°C).
12. Add Triton X-100 to 0.1% final volume, then pellet the nuclei at 600g for 5 min at 4°C.
13. Remove supernatant and resuspend in 50 ul Tagmentation mix, and place the tube at 37°C 800rpm for 30 minutes.
14. Remove tubes from thermomixer and immediately add 250 ul of Binding buffer from Qiagen minElute kit.
15. Vortex 30 seconds and transfer to minelute purification column.
16. Spin 17,000g for 1 min. Discard flowthrough and add 750ul of minelute washing buffer.
17. Spin 17,000g for 1 min. Discard flowthrough and spin 2 minutes at 17,000g to dry the column.
18. Transfer column to a new 1.5 ml tube and add 20ul of minElute Elution Buffer.
19. Incubate 2min at RT and spin at 17,000g for 1 min.
20. Discard column and keep the transposed DNA on ice until PCR reaction (can be stored at -20°C)
21. To amplify transposed DNA fragments, combine the following in a PCR tube:
    - 10ul transposed DNA
    - 2.5ul i5 primer
    - 2.5ul i7 primer
    - 25ul NEBNext 2x Mastermmix (NEB)
    - 10 ul H_2O
22. Run PCR program as follows for 5 cycles

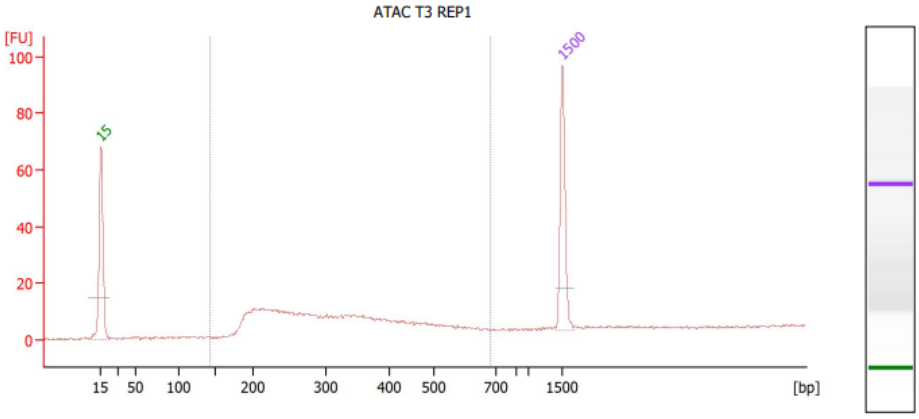


23. Remove tubes and proceed for QPCR amplification to determine the extra cycles needed by preparing the mix:
    - 3.25ul H_2O
    - 0.5ul of 25uM i5 primer
    - 0.5ul of 25uM i7 primer
    - 0.75ul Evagreen
    - 5ul NEBNcxt 2x MasterMix (NEB)
    - 5ul of Pre-Amplified Sample.
24. Then use the following program on the QPCR machine :



25. Calculate the number of cycles and run the rest of cycles needed on the PCR, then proceed to purify the PCR product
26. Use double SPRI beads size selection with 1.3x – 0.6x ratios.
    - Volume 1= Volume of starting sample x 0.6
    - Volume 2= Volume of starting sample x (1.3-0.6)
      (The starting volume is 45ul, Volume 1 = 27ul and Volume 2=31.5ul)
27. After adding V1 of beads mix by pipetting and wait 5 minutes at RT, then place on magnetic rack.
28. When the solution is clear, transfer supernatant to another tube and trash the previous magnetic beads.
29. Add V2 of beads mix and mix by pipetting and wait 5 minutes at RT. Place on magnetic rack, then trash supernatant and keep beads.
30. Add 500ul of fresh 80% ethanol without removing the magnetic rack.
31. Remove ethanol and repeat the wash.
32. Finally remove the ethanol and keep the tube open until beads are completely dry.

33. Add 20 ul of 10 mM TRIS-HCL pH 8.
34. Proceed for bioanalyzer. For tomato leaves results of bioanalyser expected to be as shown examples below.



Overall Results for sample 1 : **ATAC T3 REP1**

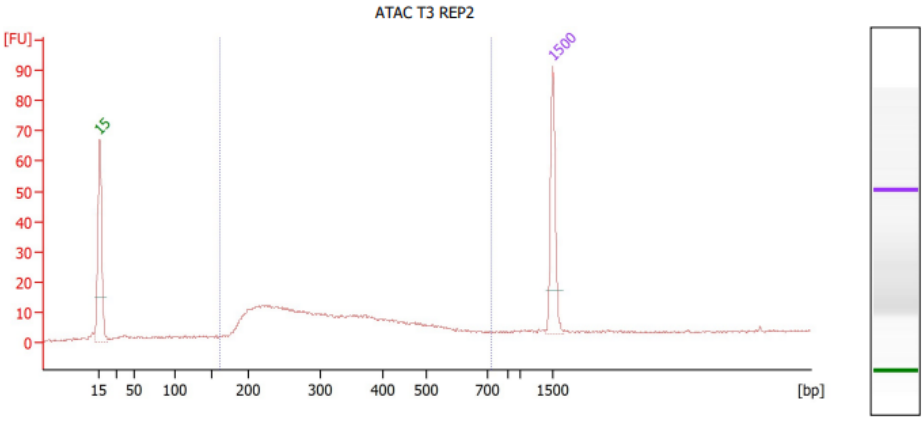
Number of peaks found:	0	Area 1:	163.0

Peak table for sample 1 : **ATAC T3 REP1**

Peak	Size [bp]	Conc. [ng/µl]	Molarity [nmol/l]	Observations
1	15	4.20	424.2	Lower Marker
2	1,500	2.10	2.1	Upper Marker

Region table for sample 1 : **ATAC T3 REP1**

From [bp]	To [bp]	% of Total	Size distribution in CV [%]	Conc. [ng/µl]	Co Area lor	Average Size [bp]	Molarity [nmol/l]
142	681	95	32.7	7.88	163.0	315	42.8



Overall Results for sample 2 : **ATAC T3 REP2**

Number of peaks found:	0	Area 1:	205.0

Peak table for sample 2 : **ATAC T3 REP2**

Peak	Size [bp]	Conc. [ng/µl]	Molarity [nmol/l]	Observations
1	15	4.20	424.2	Lower Marker
2	1,500	2.10	2.1	Upper Marker

Region table for sample 2 : **ATAC T3 REP2**

From [bp]	To [bp]	% of Total	Size distribution in CV [%]	Conc. [ng/µl]	Co Area lor	Average Size [bp]	Molarity [nmol/l]
160	730	84	34.8	10.27	205.0	325	54.9

Buffers + Mixes:

Tagmentation Mix:

- 2x Tagmentation Buffer 25ul
- 10x PBS 2ul
- 1% Digitonin 0.5ul
- 10% Tween 20 0.5ul
- H_2O 17ul
- Assembeled Trasnposomes 5ul

ATAC Lysis Buffer:

0.3M Sucrose

1.25% Ficoll

2.5% Dextran T40

15mM Tris-HCl pH 8

20mM MES

10mM $MgCl_2$

60mM KCL

15mM NaCl

0.5mM Spermine

0.5mM Spermidine

0.1% Triton X-100

H_2O (Depending on your Final Volume)

Last Minute

5mM DTT

1mM PMSF

Protease inhibitors 50ul

ATAC Washing Buffer

0.3M Sucrose

15mM Tris-HCl pH 8

15mM MES

60mM KCL

15mM NaCl

2mM EDTA

0.5mM Spermine

0.5mM Spermidine

0.1% Triton X-100

H_2O (Depending on your Final Volume)

Last Minute

5mM DTT

1mM PMSF

Protease inhibitors 50ul

ATAC Final Buffer

0.3M Sucrose

15mM Tris-HCl pH 8

15mM MES

60mM KCL

15mM NaCl

0.5mM Spermine

0.5mM Spermidine

0.1% Triton X-100

H_2O (Depending on your Final Volume)

Last Minute

5mM DTT

Protease inhibitors 50ul

Bioinformatic Analysis Pipeline for ATAC-seq library:

